Fly Climbing Assay Analyzer

Complete Undergraduate Laboratory Module

Data Collection, Analysis, and Scientific Interpretation Using Drosophila Negative Geotaxis

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Module component	Description
Student Laboratory Manual	Guided worksheet for completing the negative geotaxis analysis.
Instructor Guide	Learning outcomes, timing, preparation, and grading suggestions.
Software Tutorial	Step-by-step guide for using the Fly Climbing Assay Analyzer.
Example Dataset and Analysis	CSV export structure and R analysis workflow.
Assessment Materials	Critical-thinking questions and rubric.

Instructor Guide

Module Overview

This laboratory module introduces students to quantitative behavioral phenotyping using the Drosophila negative geotaxis assay. Students use the Fly Climbing Assay Analyzer to score climbing videos, standardize data collection, calculate locomotor performance, export analysis-ready datasets, and interpret results using statistical software. The module is intended for undergraduate genetics, physiology, neuroscience, cell biology, toxicology, aging biology, or research methods courses.

Recommended course level	Implementation options
Introductory biology lab	Use pre-recorded videos and focus on observation, scoring, and graphing.
Upper-level biology lab	Add experimental design, statistical analysis, and interpretation.
CURE or research methods course	Use class-generated data as part of an authentic research question.
Independent research training	Use as a standardized scoring and data-management workflow.

Instructional Rationale

Course-based undergraduate research experiences emphasize authentic scientific practices, including asking testable questions, collecting and analyzing data, interpreting evidence, and communicating findings. This module uses a simple behavioral assay to teach students how raw observations become quantitative datasets and how those datasets are converted into biological conclusions. The software reduces repetitive arithmetic and transcription burden so that students can focus on calibration, evidence quality, experimental variability, and interpretation.

Learning Objectives

- Explain the biological basis of the negative geotaxis response in adult Drosophila melanogaster.
- Collect standardized behavioral observations from assay videos.
- Calibrate vial boundaries and a 5-cm climbing threshold using a digital interface.
- Calculate locomotor performance as the percentage of flies reaching a defined threshold.
- Organize experimental data into a structured dataset suitable for statistical analysis.
- Generate summary statistics and graphical representations of behavioral data.
- Interpret quantitative results in relation to a biological hypothesis.
- Identify sources of experimental error and propose improvements to the assay workflow.

Estimated Time

Activity	Estimated time
Pre-lab introduction and hypothesis development	20-30 min
Video upload and software orientation	10-15 min
Calibration and scoring	30-60 min
CSV export and data cleaning	10-15 min
Graphing and statistical analysis	30-60 min
Discussion and interpretation	20-30 min

Instructor Preparation

☐ Prepare or provide negative geotaxis videos with six vials and five trials per video.

☐ Confirm that students can access the online software through GitHub Pages or by opening index.html locally.

☐ Provide an example CSV and R script if students will perform statistical analysis.

☐ Decide whether students will work individually, in pairs, or in groups.

☐ Prepare a short discussion of experimental variables such as genotype, age, sex, treatment, or exposure condition.

Student Laboratory Manual

Background

The negative geotaxis assay measures the tendency of adult Drosophila to climb upward after mechanical stimulation. Because locomotor performance can change with age, genotype, environmental exposure, or physiological condition, this assay is widely used to quantify behavioral phenotypes. In this laboratory, you will analyze negative geotaxis videos using the Fly Climbing Assay Analyzer. The software will help you standardize scoring, calculate percent climbed, and export data for analysis.

Research Question

Write the research question for your assigned experiment.

_____

_____

Hypotheses

Null hypothesis (H0):

_____

_____

Alternative hypothesis (H1):

_____

_____

Experimental Design

Design element	Student response
Independent variable	
Dependent variable	Climbing performance (%)
Control group	
Experimental group(s)	
Number of flies per vial	
Number of vials	6
Number of trials	5
Scoring time	3 seconds
Climbing threshold	5 cm

Laboratory Workflow

1. Upload assay video(s).

2. Correct video orientation if needed.

3. Set the 5-cm threshold line.

4. Set the left and right sides of each vial box.

5. Enter the total number of flies per vial.

6. Score flies that reach or exceed the 5-cm threshold for each trial.

7. Export the CSV file.

8. Import the CSV file into R, GraphPad Prism, or Excel.

9. Generate figures and perform statistical analysis.

10. Interpret the results in the context of the hypothesis.

Part I. Software Setup

☐ Open the Fly Climbing Assay Analyzer in a web browser.

☐ Upload the assigned video file(s).

☐ Verify that each video appears in its own analysis window.

☐ Use rotate or flip controls if the video is not correctly oriented.

☐ Use the enlarged viewer if needed for more accurate scoring.

Part II. Calibration

☐ Click Set 5-cm Line and select the height corresponding to 5 cm in the vial.

☐ Click Set Vial Boxes.

☐ For Vial 1, click the left side and then the right side of the vial.

☐ Repeat for Vials 2-6.

☐ Confirm that all six vial boxes are correctly displayed.

☐ Ask the instructor to check calibration before scoring.

Instructor initials: _____

Part III. Data Collection

Enter the total number of flies in each vial before scoring the trials.

Vial	Total flies
1	
2	
3	
4	
5	
6	

Trial 1

Vial	Flies >= 5 cm	% climbed
1		
2		
3		
4		
5		
6		

Trial 2

Vial	Flies >= 5 cm	% climbed
1		
2		
3		
4		
5		
6		

Trial 3

Vial	Flies >= 5 cm	% climbed
1		
2		
3		
4		
5		
6		

Trial 4

Vial	Flies >= 5 cm	% climbed
1		
2		
3		
4		
5		
6		

Trial 5

Vial	Flies >= 5 cm	% climbed
1		
2		
3		
4		
5		
6		

Part IV. Exporting Data

☐ Review all trial entries for missing values.

☐ Export the CSV file for your video or export the combined class dataset.

☐ Save the CSV using a descriptive file name.

☐ Open the CSV to confirm that the data exported correctly.

CSV file name: _____

Software Tutorial

Interface Components

Component	Purpose
Video viewer	Displays the uploaded negative geotaxis video.
Orientation controls	Rotate or flip the video for correct scoring.
Enlarged viewer	Provides a larger view while preserving orientation.
5-cm line calibration	Marks the threshold used to score climbing success.
Vial box calibration	Defines the left and right boundaries of each vial.
Total flies table	Records the total number of flies in each vial.
Trial scoring table	Records the number of flies that reached the threshold.
Results table	Displays counts and calculated percentages.
CSV export	Downloads the dataset for downstream analysis.

Recommended Student Workflow

- Start with one video and complete all calibration before scoring.
- Use the same 5-cm threshold across all trials in the video.
- Use vial boxes to prevent confusion between neighboring vials.
- Record counts only at the specified scoring time point.
- Check that count values do not exceed the total number of flies.
- Export after all five trials are complete.

Troubleshooting

Problem	Suggested solution
Video is sideways or upside down	Use rotate left, rotate right, flip horizontal, or flip vertical.
Video is hard to see	Use the enlarged viewer and pause at the scoring time.
Vial boundaries are incorrect	Clear vial boxes and re-enter left and right sides for each vial.
CSV is missing values	Return to the scoring table and check all five trials.
Percentages look wrong	Check that total flies per vial were entered correctly.

Data Analysis Module

CSV Structure

The exported CSV contains one row per vial per trial. The exact columns may vary by software version, but the dataset should include the video filename, metadata fields, trial number, vial number, total flies, count above the 5-cm threshold, and percent climbed.

Column	Meaning
VideoFile	Original video filename.
Genotype	Student-entered genotype or group.
Sex	Student-entered sex.
Treatment	Student-entered treatment or condition.
Replicate	Replicate, section, or FS number.
Trial	Trial number.
Vial	Vial number.
CountAbove5cm	Number of flies reaching the threshold.
TotalFlies	Total number of flies in that vial.
PercentClimbed	Calculated climbing performance.

Example R Workflow

Students can use the following workflow to import, summarize, and graph the exported CSV file in R.

```
library(tidyverse)

data <- read_csv("combined_fly_assay_results.csv")

summary <- data %>%
  group_by(Genotype, Sex, Treatment, Trial) %>%
  summarize(mean_percent = mean(PercentClimbed, na.rm = TRUE),
            sd_percent = sd(PercentClimbed, na.rm = TRUE),
            n = n(), .groups = "drop")

ggplot(data, aes(x = Treatment, y = PercentClimbed)) +
  geom_boxplot() +
  geom_jitter(width = 0.1, alpha = 0.6) +
  ylab("Percent climbed") +
  xlab("Treatment") +
  theme_classic()
```

Analysis Questions

1. Which group had the highest average climbing performance?
2. How much variation was observed among vials within the same group?
3. Do the results support the hypothesis?
4. What additional data would strengthen the conclusion?

Critical Thinking Questions

5. Why is it important to define the 5-cm threshold before scoring the trials?

_____

6. How might inconsistent vial boundaries influence the results?

_____

7. Why should the total number of flies be recorded for each vial?

_____

8. What sources of biological variation might affect climbing performance?

_____

9. What sources of technical variation might affect scoring accuracy?

_____

10. How does standardized software improve reproducibility?

_____

11. What features would be needed to make this tool fully automated?

_____

Student Deliverables

☐ Completed laboratory worksheet.

☐ Exported CSV file.

☐ Summary table of climbing performance.

☐ At least one figure showing locomotor performance.

☐ Statistical analysis output.

☐ Short written interpretation of results.

☐ Reflection on data quality and reproducibility.

Assessment Rubric

Criterion	Excellent (4)	Good (3)	Developing (2)	Beginning (1)
Calibration	5-cm line and vial boxes are accurate.	Minor calibration issues.	Several calibration errors.	Calibration missing or unusable.
Data collection	All counts complete and plausible.	Minor missing or corrected values.	Multiple missing values.	Data incomplete.
Data organization	CSV exported and labeled clearly.	CSV usable with minor issues.	Dataset requires substantial cleanup.	Dataset not usable.
Analysis	Appropriate graph and statistical test.	Mostly appropriate analysis.	Limited or incomplete analysis.	No meaningful analysis.
Interpretation	Conclusion clearly tied to evidence.	Conclusion mostly supported.	Conclusion weakly supported.	Conclusion unsupported.
Reflection	Thoughtful discussion of error and improvement.	Some discussion of limitations.	Limited reflection.	No reflection.

Instructor Notes and Answer Guide

Suggested Pre-lab Discussion

- What does negative geotaxis measure?
- Why is a standardized scoring time important?
- Why do we calculate percentages instead of using raw counts only?
- What makes behavioral data variable?
- How can software reduce data-entry errors?

Expected Student Outcomes

Students should be able to explain how raw video observations are transformed into quantitative behavioral measurements. Strong student responses should connect calibration, accurate counting, percentage calculation, statistical analysis, and biological interpretation.

Instructor Implementation Options

Option	Description
Short activity	Use a single video and focus on calibration, scoring, and percent calculation.
Full laboratory	Use multiple groups or treatments and include statistical analysis.
CURE-style project	Have students generate videos from ongoing research questions and contribute to a class dataset.
Computational extension	Ask students to propose future automated detection methods.

Accessibility and Inclusion Notes

The module can be completed with prerecorded videos, allowing students who cannot directly participate in animal handling or recording to contribute to data analysis. Students may work in pairs so that one student controls the software while another checks observations. Instructors should provide captions, high-contrast screenshots, and alternative written descriptions when possible.

References for Instructor Context

Auchincloss LC, et al. 2014. Assessment of course-based undergraduate research experiences: A meeting report. CBE-Life Sciences Education 13:29-40.

Corwin LA, Graham MJ, Dolan EL. 2015. Modeling course-based undergraduate research experiences: An agenda for future research. CBE-Life Sciences Education 14:es1.

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Nichols CD, Becnel J, Pandey UB. 2012. Methods to assay Drosophila behavior. Journal of Visualized Experiments 61:e3795.

R Core Team. R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing.